25CS303 OBJECT ORIENTED PROGRAMMING

2 0 2 3

CLASSES AND OBJECT-ORIENTED CONCEPTS: Object oriented programming paradigms – Java buzzwords – Data types, variables and arrays – Operators – Control statements – Programming structures in Java – Classes – Constructors – Methods – Access specifiers – Static members –- Overloading methods - Objects as parameters - Returning objects. **(6)**

INHERITANCE, PACKAGES AND INTERFACES: Inheritance Basics – Types of inheritance – Super keyword - Method overriding - Dynamic method dispatch - Abstract classes - Final keyword with inheritance – Packages - Packages and member access - Importing packages – Interfaces. **(6)**

EXCEPTION HANDLING AND MULTITHREADING: Exception Handling basics – Multiple catch clauses – Nested try statements – Java's built-in exceptions – User defined exception. Multithreaded programming: Java thread model – creating a thread and multiple threads – Priorities – Synchronization – Inter thread communication-Suspending – Resuming and Stopping threads. . **(6)**

I/O AND STRING HANDLING: I/O Basics – Reading and Writing console I/O – Reading and writing files. Strings: Basic String class, methods and String buffer class – Wrappers – Auto boxing. **(6)**

GENERICS AND COLLECTIONS: Generics: Generic programming – Generic classes – Generic Methods – Bounded Types – Restrictions and Limitations. Collection Framework in Java: List – Array list – Hash table – Hash set – Tree set – Stack – Priority queue. **(6)**

LIST OF EXPERIMENTS

1. Classes and Methods
2. Inheritance
3. Abstract Classes
4. Interfaces and Packages
5. Exception Handling
6. Multithreading Concepts
7. I/O and Files
8. Generics and Lambda Expressions
9. Collection Classes

Total L: 30 + P: 30 = 60 periods

TEXTBOOKS:

1. Herbert Schildt, *'Java: The Complete Reference'*. 11th edition, McGraw Hill Education, New Delhi, 2019.
2. Deitel P and Deitel H, "*Java: How to Program*". 11th edition, Prentice Hall, 2018.

REFERENCES:

1. Cay S. Horstmann, *'Core Java Fundamentals'*. Volume 1, Eleventh Edition, Prentice Hall, 2018.
2. James Gosling, Bill Joy, Guy Steele, Gilad Bracha, Alex Buckley and Daniel Smith, "*The Java Language Specification – Java SE*". Thirteenth Edition, Oracle America Inc., USA, 2019.

COURSE OUTCOMES

At the end of the course, students will be able to:		**Bloom's Level**
CO1	Explain the core Object Oriented Programming concepts and their applications.	**K2**
CO2	Apply Java constructs to develop various applications.	**K3**
CO3	Analyze solutions involving code reusability and complexity management	**K4**
CO4	Develop modular and reusable applications by integrating OOP principles and advanced programming concepts.	
CO5	Demonstrate problem-solving skills through the development of well-structured and maintainable applications.	

COs-POs & PSOs MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1												2	2
CO2	3											3	3
CO3		1										1	1

CO4					3			3				2	2
CO5					1			1				1	1
@	**3**	**1**			**3**			**3**				**3**	**3**

1-low, 2-medium, 3-high @-Overall Contribution to the Course

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1												**2**	**2**
CO2	**3**											**3**	**3**
CO3		**1**										**1**	**1**
CO4					**3**			**3**				**3**	**3**
CO5					**1**			**1**				**1**	**1**
	3	**1**			**3**			**3**				**3**	**3**

1-low,2-medium,3-high

Assessment Pattern:

	CO1	CO2	CO3	CO4	CO5	Total
CAT1	15	35	10			60
CAT2	20	30	10			60
Assignment		30	10			40
CCAT	10	20	10			40
Lab				80	10	90
Total	**45**	**115**	**40**	**80**	**10**	**290**

Assessment Pattern:

CO	Percentage	Weightage
CO1	16%	2
CO2	40%	3
CO3	14%	1
CO4	28%	3
CO5	3%	1

Above 25% - 3

15%- 25% - 2

Below 15% - 1